

TOTAL PICKS

166

HR HITS

58

HIT RATE

34.9%

HIT STREAK

21d

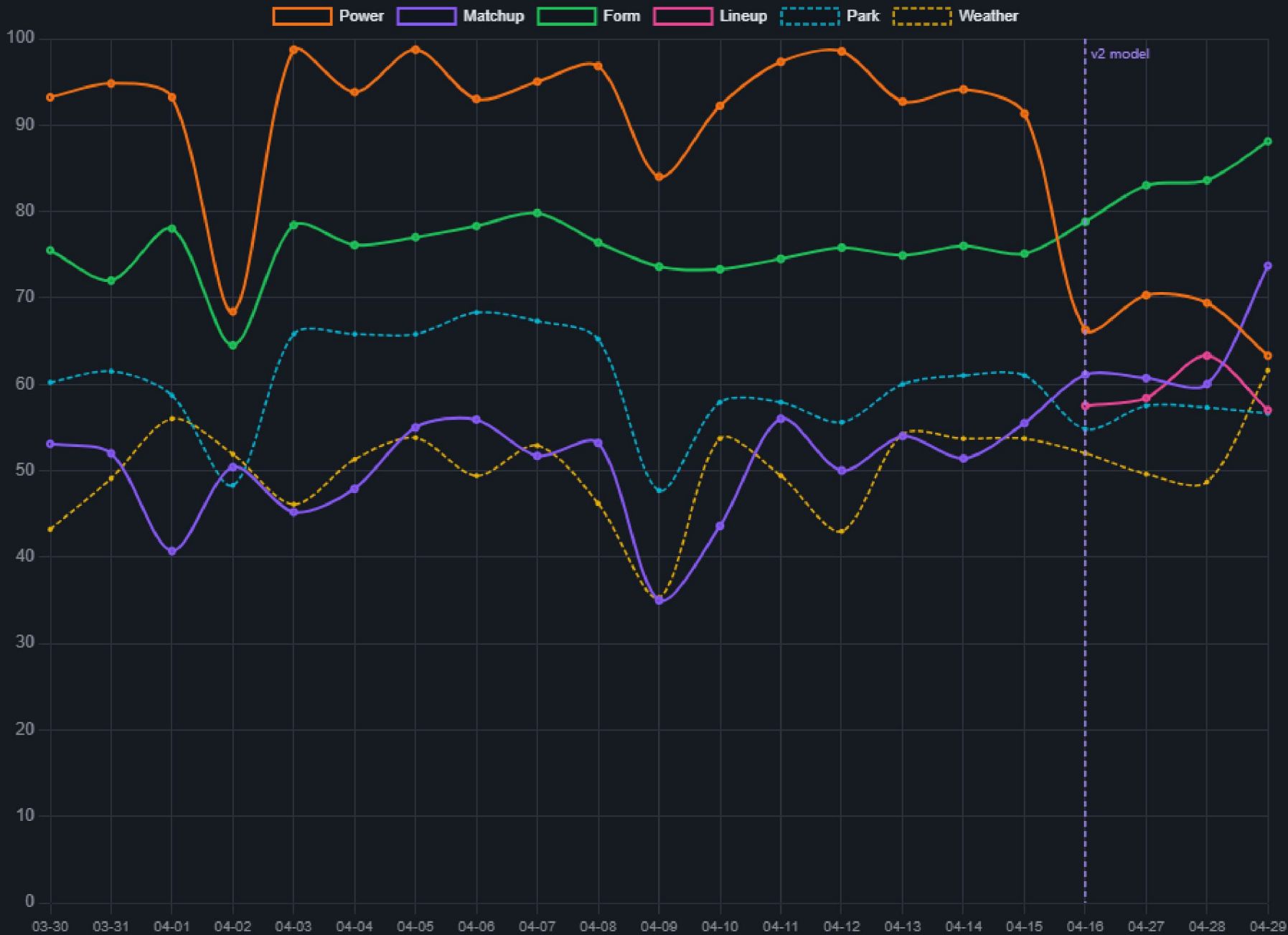
AVG COMP (HITS)

69.8

AVG COMP (MISS)

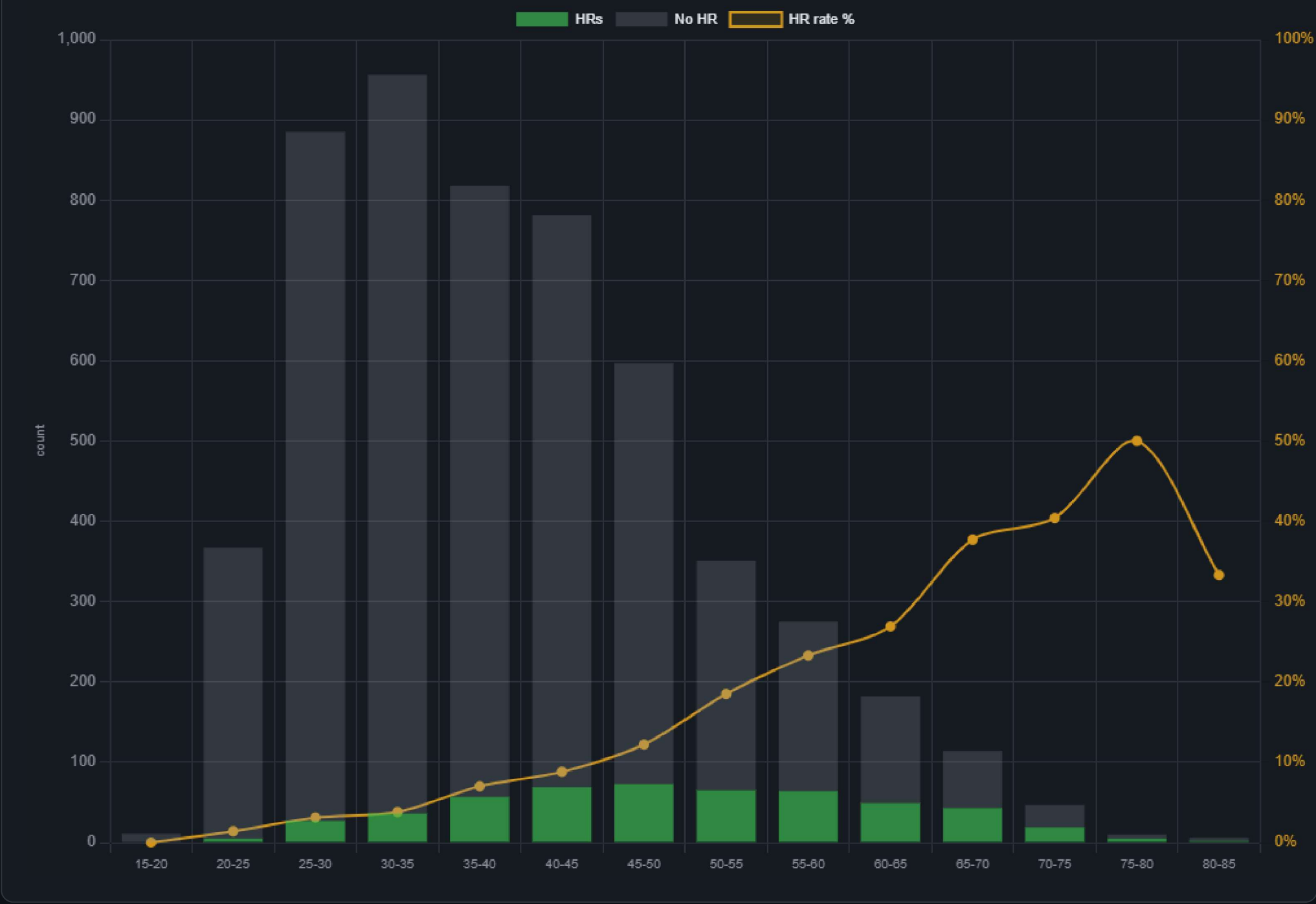
67.8

Factor Score Trends (Selected Picks)



Composite Score → HR Rate (full board, all dates)

Bars: counts of HRs (green) and non-HRs (gray) per composite range. Orange line: HR rate % (right axis) — the model's signal climbs from low single digits at the bottom to ~40-50% at the top.



Per-Factor Mean — HR Hitters vs Misses (last 30 days)

Mean factor score for batters who hit a HR vs those who didn't. A bigger green-vs-gray gap = the factor is surfacing HR hitters. A flat pair = the factor isn't differentiating and is a candidate to drop or rework.



POWER +22.2 MATCHUP +6.9 PARK +0.9 FORM +16.6 WEATHER +2.5 LINEUP +0

Gaps (HR mean – Miss mean) over last 30 days. Composite gap: +11.3 on 417 HR-days vs 3986 non-HR. A factor with gap < 3 is providing little signal.

Factor Decomposition — What's Inside Each Score?

For each factor, the underlying raw inputs the model is consuming. Green = HR-hitter mean, gray = miss mean. The All view shows overall signal; band views show what differs INSIDE a composite range (e.g., "of guys we scored 40-60, what do the HR hitters share?").

Composite band:

All0-4040-6060-8080+

Daily rank band:

#1-10#11-30#31-100#101-300

Sources: 162 live rows, 4425 backfilled (current-stats proxy for older days).

Sample: 417 HR-batter rows / 3986 miss-batter rows over last 30 days.

POWER

INPUT	HR MEAN	MISS MEAN	GAP	VISUAL
barrel_pct	8.75	5.10	+3.65	
exit_velo	88.89	87.69	+1.19	
hr_fb_pct	7.90	4.59	+3.30	
iso	0.20	0.14	+0.06	
xwoba_contact	0.35	0.32	+0.03	
pull_fb_pct			no data yet	

MATCHUP

INPUT	HR MEAN	MISS MEAN	GAP	VISUAL
pitcher_hr_per_9	1.42	1.10	+0.32	
pitcher_era	4.67	4.17	+0.50	
pitcher_hh_pct	32.74	31.57	+1.17	
pitcher_k_per_9	8.26	8.58	-0.32	
pitcher_fb_pct_allowed	93.42	93.25	+0.17	
woba_vs_hand	0.38	0.34	+0.04	
archetype_similarity	74.66	72.18	+2.48	
vegas_implied_total			no data yet	
platoon_advantage	0.00	0.00	0.00	

FORM

INPUT	HR MEAN	MISS MEAN	GAP	VISUAL
recent_hr_14d	1.21	0.89	+0.32	
recent_barrel_pct_14d			no data yet	
ev_trend_14d			no data yet	

PARK

INPUT	HR MEAN	MISS MEAN	GAP	VISUAL
hr_park_factor	105.06	102.83	+2.23	

WEATHER

INPUT	HR MEAN	MISS MEAN	GAP	VISUAL
temperature_f	66.49	63.10	+3.39	
wind_mph	7.71	7.10	+0.61	
humidity_pct	64.00	65.06	-1.06	
is_dome	0.23	0.23	+0.01	

LINEUP

INPUT	HR MEAN	MISS MEAN	GAP	VISUAL
batting_order	4.25	4.92	-0.67	

Why Are HR Hitters Stuck at 40-60?

Same inputs as above, but compares **HR hitters who scored 40-60** (model under-rated them) vs **HR hitters who scored 60-80** (model caught them). The biggest input gaps here are the signals the model fails to convert into score for the under-rated group — candidates for weight increase or new feature.

Sample: 216 HR hitters scored 40-60 vs 100 HR hitters scored 60-80.

POWER

INPUT	HR HITTERS SCORED 40-60	HR HITTERS SCORED 60-80	Δ (LOW-HIGH)
barrel_pct	8.38	12.98	-4.60

INPUT	HR HITTERS SCORED 40-60	HR HITTERS SCORED 60-80	Δ (LOW–HIGH)
exit_velo	88.78	90.40	-1.62
hr_fb_pct	7.55	11.75	-4.20
iso	0.20	0.29	-0.09
xwoba_contact	0.34	0.39	-0.05
pull_fb_pct	—	—	—
MATCHUP			
INPUT	HR HITTERS SCORED 40-60	HR HITTERS SCORED 60-80	Δ (LOW–HIGH)
pitcher_hr_per_9	1.39	1.72	-0.34
pitcher_era	4.56	5.32	-0.76
pitcher_hh_pct	32.37	34.34	-1.97
pitcher_k_per_9	8.27	7.89	+0.39
pitcher_fb_pct_allowed	93.26	94.06	-0.80
woba_vs_hand	0.37	0.43	-0.06
archetype_similarity	74.28	74.24	+0.04
vegas_implied_total	—	—	—
platoon_advantage	0.00	0.00	0.00
FORM			
INPUT	HR HITTERS SCORED 40-60	HR HITTERS SCORED 60-80	Δ (LOW–HIGH)
recent_hr_14d	1.09	2.31	-1.22
recent_barrel_pct_14d	—	—	—
ev_trend_14d	—	—	—
PARK			
INPUT	HR HITTERS SCORED 40-60	HR HITTERS SCORED 60-80	Δ (LOW–HIGH)
hr_park_factor	103.60	107.17	-3.57
WEATHER			
INPUT	HR HITTERS SCORED 40-60	HR HITTERS SCORED 60-80	Δ (LOW–HIGH)
temperature_f	63.59	72.07	-8.48
wind_mph	8.91	5.22	+3.69
humidity_pct	65.22	66.67	-1.45
is_dome	0.10	0.50	-0.40
LINEUP			
INPUT	HR HITTERS SCORED 40-60	HR HITTERS SCORED 60-80	Δ (LOW–HIGH)
batting_order	4.67	3.83	+0.83

HR Hitters by Daily Rank — Where's the Tunable Cohort?

Daily rank controls for slate variance. Compares **HR hitters we picked (#1-10)** vs **HR hitters we just missed (#11-30)** vs **deep-board HR hitters (#31-100)**. The largest input gaps between #1-10 and #11-30 are the most actionable tuning levers — these are the closest cohort to breaking into the top-8.

Sample: **63** HR hitters at HR hitters ranked #1-10, **85** at HR hitters ranked #11-30, **135** at HR hitters ranked #31-100. Δ shows fringe minus top — a negative gap means rank-#11-30 HR hitters were systematically weaker on that input.

POWER

INPUT	HR HITTERS RANKED #1-10	HR HITTERS RANKED #11-30	HR HITTERS RANKED #31-100	Δ (FRINGE–TOP)
barrel_pct	13.51	10.72	8.22	-2.79
exit_velo	90.74	89.52	88.73	-1.22
hr_fb_pct	12.24	9.69	7.40	-2.55
iso	0.30	0.24	0.19	-0.06

CONTACT				
INPUT	HR HITTERS RANKED #1-10	HR HITTERS RANKED #11-30	HR HITTERS RANKED #31-100	Δ (FRINGE–TOP)
xwoba_contact	0.40	0.36	0.34	-0.04
pull_fb_pct	—	—	—	—
MATCHUP				
INPUT	HR HITTERS RANKED #1-10	HR HITTERS RANKED #11-30	HR HITTERS RANKED #31-100	Δ (FRINGE–TOP)
pitcher_hr_per_9	1.89	1.45	1.41	-0.44
pitcher_era	5.50	4.74	4.66	-0.75
pitcher_hh_pct	34.56	32.96	32.58	-1.60
pitcher_k_per_9	7.85	8.27	8.20	+0.42
pitcher_fb_pct_allowed	94.21	93.58	93.28	-0.63
woba_vs_hand	0.44	0.39	0.37	-0.05
archetype_similarity	73.97	74.61	73.77	+0.64
vegas_implied_total	—	—	—	—
platoon_advantage	0.00	0.00	0.00	0.00
FORM				
INPUT	HR HITTERS RANKED #1-10	HR HITTERS RANKED #11-30	HR HITTERS RANKED #31-100	Δ (FRINGE–TOP)
recent_hr_14d	2.29	1.92	1.03	-0.37
recent_barrel_pct_14d	—	—	—	—
ev_trend_14d	—	—	—	—
PARK				
INPUT	HR HITTERS RANKED #1-10	HR HITTERS RANKED #11-30	HR HITTERS RANKED #31-100	Δ (FRINGE–TOP)
hr_park_factor	107.00	107.20	105.86	+0.20
WEATHER				
INPUT	HR HITTERS RANKED #1-10	HR HITTERS RANKED #11-30	HR HITTERS RANKED #31-100	Δ (FRINGE–TOP)
temperature_f	72.00	72.08	62.79	+0.08
wind_mph	0.00	6.26	10.03	+6.26
humidity_pct	—	66.67	67.43	—
is_dome	1.00	0.40	0.00	-0.60
LINEUP				
INPUT	HR HITTERS RANKED #1-10	HR HITTERS RANKED #11-30	HR HITTERS RANKED #31-100	Δ (FRINGE–TOP)
batting_order	3.00	4.00	4.00	+1.00

Input Calibration — Score Curve vs Empirical HR Rate

For each raw input, we bin into 5 quantile buckets and compare the empirical HR rate per bucket to the average sub-factor score we assigned. **Aligned** = the score curve climbs with HR rate (correct). **Backwards** = the score climbs while HR rate drops or vice versa. **Flat** = no signal in either curve. The most actionable diagnostic for fixing per-input score functions.

POWER

barrel_pct					PARTIAL	
BIN	RANGE	N	HR RATE %	AVG POWER SCORE	CURVES	
1	0–1.7	1080	0.6%	14.4		
2	1.7–3.8	1080	5.3%	10.6		
3	3.8–5.8	1080	7.9%	19		
4	5.8–9.1	1080	12.6%	32.9		
5	9.1–25	1080	21.2%	57.2		
exit_velo					PARTIAL	

	BIN	RANGE	N	HR RATE %	AVG POWER SCORE	CURVES
	1	82–86.5	1080	2.6%	16.4	
	2	86.5–87.4	1080	6%	12.8	
	3	87.4–88.1	1080	7.8%	18	
	4	88.1–89.1	1080	11.5%	28.6	
	5	89.1–97	1080	19.7%	58.2	

hr_fb_pct

PARTIAL

	BIN	RANGE	N	HR RATE %	AVG POWER SCORE	CURVES
	1	0–1.5	1080	0.6%	14.5	
	2	1.5–3.5	1080	5.5%	10.3	
	3	3.5–5.2	1080	7.8%	19.1	
	4	5.2–8.2	1080	12.6%	32.9	
	5	8.2–36	1080	21.2%	57.2	

iso

PARTIAL

	BIN	RANGE	N	HR RATE %	AVG POWER SCORE	CURVES
	1	0–0.083	1080	1.2%	15.9	
	2	0.084–0.122	1080	5.8%	8.6	
	3	0.122–0.155	1080	7.9%	16.9	
	4	0.155–0.214	1080	12.4%	33.6	
	5	0.214–0.6	1080	20.3%	59.1	

xwoba_contact

PARTIAL

	BIN	RANGE	N	HR RATE %	AVG POWER SCORE	CURVES
	1	0–0.277	1080	4.7%	18.6	
	2	0.277–0.301	1080	5.5%	13.5	
	3	0.301–0.33	1080	10.1%	26.1	
	4	0.33–0.365	1080	11.3%	29.3	
	5	0.365–1.151	1080	16%	46.5	

MATCHUP

pitcher_fb_pct_allowed

BACKWARDS

	BIN	RANGE	N	HR RATE %	AVG MATCHUP SCORE	CURVES
	1	86.8–91.6	1040	9.3%	37.4	
	2	91.6–92.9	1040	7.8%	39.2	
	3	92.9–94	1040	10%	39.6	
	4	94–94.8	1040	8.6%	41.1	
	5	94.8–98.9	1040	11.4%	44.1	

archetype_similarity

BACKWARDS

	BIN	RANGE	N	HR RATE %	AVG MATCHUP SCORE	CURVES
	1	44.9–62.8	1080	5.8%	44.2	
	2	62.9–70.1	1080	9.1%	41.9	
	3	70.1–77	1080	10%	38.2	
	4	77–81.6	1080	11.2%	38.6	
	5	81.6–94.1	1080	11.5%	40	

pitcher_hr_per_9

PARTIAL

	BIN	RANGE	N	HR RATE %	AVG MATCHUP SCORE	CURVES
	1	0–0.59	1033	5.1%	27.6	
	2	0.59–0.84	1033	7.7%	34.8	
	3	0.84–1.27	1032	8%	37.8	
	4	1.27–1.55	1032	10.8%	45.7	
	5	1.55–5.07	1032	15.3%	54.8	

pitcher_era

PARTIAL

	BIN	RANGE	N	HR RATE %	AVG MATCHUP SCORE	CURVES
	1	0.6–2.7	1033	7.8%	34.6	
	2	2.7–3.21	1033	7.7%	35.1	
	3	3.21–4.3	1032	7.8%	36.9	
	4	4.3–5.84	1032	9.4%	43.4	
	5	5.84–12.54	1032	14.1%	50.7	

pitcher_hh_pct

PARTIAL

	BIN	RANGE	N	HR RATE %	AVG MATCHUP SCORE	CURVES
	1	25–26	1033	8.2%	36	
	2	26–28.6	1033	8%	34.8	
	3	28.6–32	1032	7.9%	38.7	
	4	32–36.6	1032	10.6%	45.3	
	5	36.6–50	1032	12.2%	45.9	

pitcher_k_per_9						PARTIAL
	BIN	RANGE	N	HR RATE %	AVG MATCHUP SCORE	CURVES
	1	3.96–6.32	1033	11.9%	46.8	
	2	6.32–7.68	1033	8.8%	38.5	
	3	7.68–9.3	1032	9.4%	39.2	
	4	9.3–10.2	1032	8.3%	39.5	
	5	10.29–16.36	1032	8.5%	36.6	

woba_vs_hand						PARTIAL
	BIN	RANGE	N	HR RATE %	AVG MATCHUP SCORE	CURVES
	1	0–0.283	1080	3.8%	39.8	
	2	0.283–0.325	1080	6.3%	40.2	
	3	0.325–0.354	1080	9.2%	40.5	
	4	0.354–0.394	1080	11.2%	40.8	
	5	0.394–0.65	1080	17.1%	41.6	

FORM

recent_hr_14d						PARTIAL
	BIN	RANGE	N	HR RATE %	AVG FORM SCORE	CURVES
	1	0–0	1080	8.5%	35.6	
	2	0–0	1080	5.8%	29.4	
	3	0–1	1080	10.2%	32.8	
	4	1–2	1080	10%	41.7	
	5	2–7	1080	13.1%	55.7	

PARK

hr_park_factor						PARTIAL
	BIN	RANGE	N	HR RATE %	AVG PARK SCORE	CURVES
	1	92–97	29	6.9%	11.2	
	2	97–101	28	7.1%	29.1	
	3	101–105	28	14.3%	46.1	
	4	105–107	28	14.3%	66.2	
	5	107–118	28	17.9%	89.1	

WEATHER

wind_mph						BACKWARDS
	BIN	RANGE	N	HR RATE %	AVG WEATHER SCORE	CURVES
	1	0–0	29	10.3%	50	
	2	0–7.9	28	10.7%	56.6	
	3	7.9–9.3	28	7.1%	58.9	
	4	9.3–9.7	28	14.3%	45.9	
	5	9.7–12.8	28	17.9%	47.6	

humidity_pct						BACKWARDS
	BIN	RANGE	N	HR RATE %	AVG WEATHER SCORE	CURVES
	1	36–36	22	4.5%	50	
	2	36–64	22	13.6%	60.8	
	3	64–68	22	31.8%	57.1	
	4	68–88	21	4.8%	58.4	
	5	88–99	21	4.8%	34.7	

temperature_f						ALIGNED
	BIN	RANGE	N	HR RATE %	AVG WEATHER SCORE	CURVES
	1	48.6–50.2	29	3.4%	34.4	
	2	50.2–59.4	28	10.7%	44.4	
	3	59.4–72	28	17.9%	62.3	
	4	72–72	28	10.7%	50	
	5	72–76.6	28	17.9%	68.4	

Dome vs Outdoor — Is Our Dome Bias Justified?

Dome games get a flat weather score (50, neutral). Outdoor games can score higher with helping wind but lower with hostile wind. If domes convert at a higher HR rate than outdoor games of comparable park-factor, the bias is justified. If not, we're under-rewarding outdoor wind alignment.

DOMES SHARE OF PICKS

0.6%

1 / 166

DOMES HR RATE

100%

OUTDOOR HR RATE

34.5%

BIAS SIGNAL

Dome bias justified

Δ 65.5 pts

DOMES	PARK FACTOR BAND	N	HR RATE	AVG WEATHER SCORE	SELECTED PICKS
 Dome	105-110 (hitter)	18	22.2%	50	1
 Dome	110+ (strong hitter)	15	0%	50	0
 Outdoor	100-105 (mild hitter)	39	10.3%	46.3	2
 Outdoor	105-110 (hitter)	13	15.4%	68.8	0
 Outdoor	110+ (strong hitter)	11	27.3%	57.8	2
 Outdoor	95-100 (mild pitcher)	32	9.4%	44.6	1
 Outdoor	<95 (pitcher)	13	7.7%	68.4	2
 Outdoor	unknown	5	0%	59.6	0
 Outdoor	unknown	5254	9.5%	46.2	158

Wind Direction Effect — Outdoor Games Only

HR rate by wind-helping band (cosine of wind-to-CF angle × MPH). Out-blowing wind toward CF should produce higher HR rates than in-blowing wind toward home plate. The avg_weather_score column shows what our model is rewarding for each band — if HR rate climbs but score is flat, the wind effect isn't being weighted enough.

Sample: 108 outdoor pick-rows over last 60 days. Venues missing CF bearing: Sutter Health Park

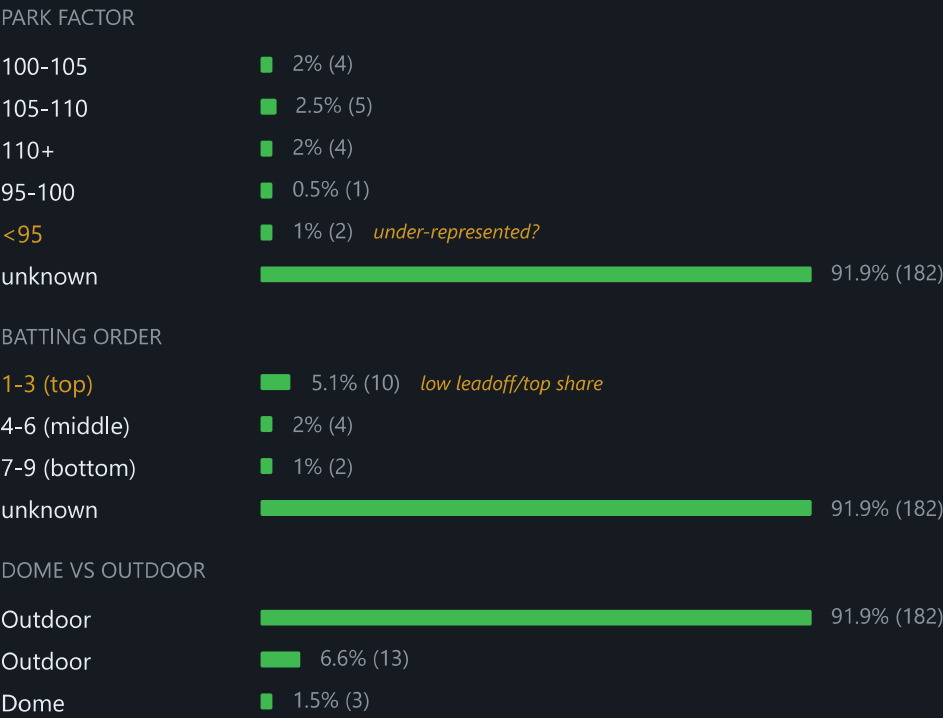
WIND BAND	N	HR RATE %	AVG HELPING MPH	AVG WEATHER SCORE
Wind IN (5+ MPH)	47	6.4%	-8.7	35.8
Wind IN (1-5 MPH)	37	21.6%	-2.9	66.5
Wind OUT (1-5 MPH)	24	8.3%	3.5	62.8

Outdoor games only. helping_factor = wind_mph × cos(angle_to_CF). Score column shown is weather_score (model output). If HR rate climbs out_strong→in_strong but score is flat, the wind effect isn't being rewarded enough.

Pick Composition — What Does Our Top-8 Look Like in Aggregate?

Distribution of selected picks by park factor, batting order, and dome status. Surfaces systematic biases that aren't visible per-pick — e.g., 80% dome share, never picks from sub-95 park-factor venues, leadoff hitters absent.

Total selected picks in window: 198



Top Hitters (3+ picks)

1	Jordan Walker	STL	7/15	47%	68.6
2	Sal Stewart	CIN	6/13	46%	69.1
3	Yordan Alvarez	HOU	5/15	33%	71.9

4	Gary Sánchez	MIL	4/8	50%	71.3
5	CJ Abrams	WSH	4/13	31%	69.9
6	Aaron Judge	NYY	4/8	50%	67.9
7	Ben Rice	NYY	4/11	36%	66.7
8	Mickey Moniak	COL	3/9	33%	76.6
9	Brandon Lowe	PIT	3/7	43%	70.8
10	Matt Olson	ATL	3/9	33%	67.5
11	Dalton Rushing	LAD	2/3	67%	71.4
12	Kyle Schwarber	PHI	2/6	33%	71
13	Mike Trout	LAA	2/4	50%	65.8
14	Gunnar Henderson	BAL	1/7	14%	67.8
15	Elly De La Cruz	CIN	1/6	17%	65.6